

USE CASE DOCUMENT

AutoGen Multi-Agent Tax Filing Assistant

An Agentic AI System for Indian Income Tax Computation, Advisory & Compliance (FY 2024-25)

Framework: Microsoft AutoGen / AG2 (LLM-orchestrated multi-agent framework)

Domain: Taxation / Personal Finance / GenAI Solution Engineering

Document type: Solution Use Case & Technical Walkthrough

1. Executive Summary

This document describes a working agentic-AI use case implemented with the Microsoft AutoGen (AG2) framework. The application functions as an automated Income Tax Filing Assistant for Indian taxpayers for the Financial Year 2024-25. It demonstrates how multiple specialised Large Language Model (LLM) agents can collaborate — each owning a distinct responsibility — to take a taxpayer's raw financial details and return a complete, reasoned filing outcome: tax computed under both regimes, applicable deductions, the recommended ITR form, and a data-validation report.

Rather than relying on a single monolithic prompt, the solution decomposes the tax-filing problem into discrete, tool-backed agents that converse in a managed group chat. A central manager decides which agent speaks next based on the state of the conversation, and each agent invokes deterministic Python functions for any calculation that must be accurate and auditable. This hybrid design — LLM reasoning for orchestration and natural-language interaction, deterministic code for numeric correctness — is the central architectural idea the use case illustrates.

2. Business Context & Problem Statement

Individual income-tax filing in India is procedurally complex. A taxpayer must choose between the old and new tax regimes, identify which deductions and exemptions apply to their profile, select the correct ITR form, and ensure mandatory data (PAN, Aadhaar linkage, pre-validated bank account) is complete before filing. Errors lead to rejected returns, notices, or missed savings.

The traditional options are either expensive professional assistance or generic calculators that handle one slice of the problem in isolation. The opportunity this use case addresses is to provide an end-to-end, conversational assistant that reasons about a taxpayer's complete profile and produces a coherent, validated filing recommendation in a single interaction.

Challenge	Description
Regime selection	Old vs. new regime trade-offs depend on the individual's deduction profile and require computing both before comparing.
Deduction discovery	Eligible sections (80C, 80D, 24(b), 80CCD(1B), HRA, 80G) vary by taxpayer and are easily overlooked.
Form selection	The correct ITR form depends on income type, total income, and entity status.
Pre-filing validation	Missing PAN, unlinked Aadhaar, or an unvalidated bank account silently block filing or refunds.

3. Solution Overview

The solution is a multi-agent system in which four cooperating agents and a group-chat manager process the taxpayer scenario. The taxpayer provides their details once as a natural-language message; the agents then collaborate, calling registered Python tools where deterministic computation is needed, and converge on a consolidated answer.

3.1 What the AutoGen Framework Provides

AutoGen is an open-source framework for building collaborative, multi-agent AI systems in which LLM-driven agents communicate to solve complex tasks autonomously. Architecturally it provides three building blocks that this use case relies on:

- **User Proxy Agent** — represents the human or system initiating the request and executes registered tool functions.
- **Assistant Agents** — the reasoning and problem-solving agents that interpret the request and decide which tools to call.
- **Tool / Function layer** — the deterministic Python functions registered to agents for accurate calculation and data retrieval.

4. Actors & Agent Roles

The system is composed of one user-side proxy, three assistant agents, and an orchestration manager. Each assistant agent owns a single responsibility and is backed by one or more registered tool functions.

Agent	Type	Responsibility	Tool(s) bound
TaxpayerProxy	User Proxy	Represents the taxpayer; collects income and deduction inputs and executes the registered tool functions on behalf of the agents.	(executor for all tools)
TaxComputationAgent	Assistant	Computes income-tax liability under the FY 2024-25 slabs.	compute_india_tax
TaxAdvisoryAgent	Assistant	Suggests applicable deductions, exemptions and saving strategies based on the taxpayer profile.	suggest_deductions
ComplianceAgent	Assistant	Validates mandatory data, flags errors/warnings, and recommends the correct ITR form.	validate_tax_data, recommend_itr_form
GroupChatManager	Orchestrator	Drives the conversation, routing messages and selecting which agent speaks next.	(LLM auto-selection)

5. Functional Components (Registered Tools)

Four deterministic Python functions are registered as callable tools. Keeping these computations in code rather than asking the LLM to do arithmetic guarantees numeric correctness and auditability.

5.1 `compute_india_tax(gross_income, regime)`

Computes income tax for FY 2024-25 under either the new (default) or old regime. The logic captures:

- **Standard deduction** — ₹75,000 under the new regime, ₹50,000 under the old regime.
- **Slab computation** — progressive slab rates specific to each regime (new regime: nil up to ₹3L, then 5% / 10% / 15% / 20% / 30%; old regime: nil up to ₹2.5L, then 5% / 20% / 30%).
- **Rebate** — Section 87A rebate of up to ₹25,000 where taxable income under the new regime is ≤ ₹7L.
- **Surcharge** — 10% above ₹50L taxable and 15% above ₹1Cr.
- **Cess** — Health & Education Cess at 4% on tax plus surcharge.

It returns a structured dictionary including taxable income, base tax, rebate, surcharge, cess, total tax payable, and the effective tax rate.

5.2 `suggest_deductions(income_details)`

Inspects a taxpayer profile (flags such as `has_80C_investments`, `has_health_insurance`, `has_home_loan`, `has_nps`, `has_hra`, `donations`) and returns the applicable deduction sections with their statutory ceilings — 80C (₹1.5L), 80D (₹75K), 24(b) home-loan interest (₹2L), 80CCD(1B) NPS (₹50K), HRA exemption, and 80G donations — together with an estimate of additional savings. It notes that most of these apply primarily under the old regime, while 80CCD(1B) is also available under the new regime.

5.3 `recommend_itr_form(income_type, total_income, is_company)`

Maps the taxpayer's income type and total income to the correct ITR form — ITR-1 (Sahaj) for salaried individuals up to ₹50L, ITR-2 for higher income or capital gains, ITR-3 for business/professional income, ITR-4 (Sugam) for presumptive income, and ITR-6 for companies — and returns the reason, due dates, and the official e-filing portal URL.

5.4 `validate_tax_data(data)`

Performs pre-filing validation: PAN presence and 10-character format, Aadhaar linkage, non-negative gross income, a TDS-vs-income sanity check against Form 26AS, and the presence of a pre-validated bank account for refunds. It returns a validity flag with categorised errors and warnings and a recommended action.

6. End-to-End Workflow

The interaction proceeds as an orchestrated group chat. A single taxpayer scenario kicks off the session and the manager routes the conversation across agents until the task is complete.

1. **Input** — The taxpayer submits a natural-language scenario containing income, investments, and a list of requested outcomes.
2. **Orchestration** — The GroupChatManager selects the next speaker automatically based on conversation state (auto speaker selection, up to 15 rounds).
3. **Computation** — The TaxComputationAgent calls `compute_india_tax` for both regimes; the TaxpayerProxy executes the function and returns the figures.
4. **Advisory** — The TaxAdvisoryAgent identifies eligible deductions via `suggest_deductions`.
5. **Compliance** — The ComplianceAgent validates the data and recommends the appropriate ITR form.
6. **Consolidation** — The agents converge on a consolidated recommendation — best regime, deductions, ITR form, and an e-filing readiness report — captured in the conversation history.

The notebook ships with a representative taxpayer scenario used to drive the session:

Field	Value
Name	Rahul Sharma (Age 35, Resident Individual)
PAN / Aadhaar	ABCPS1234R / Linked
Gross Salary	₹18,00,000
HRA received	₹2,40,000 (rent ₹25,000 / month)
Savings interest	₹12,000
TDS deducted	₹1,85,000
Investments	PPF ₹50K, ELSS ₹50K, LIC ₹25K, Health insurance ₹22K, Home-loan interest ₹1.8L, NPS Tier-1 ₹50K
Requested outcomes	Tax under both regimes; applicable deductions; best regime & ITR form; data validation and e-filing guidance

8. Technical Implementation Notes

Aspect	Detail
Framework	Microsoft AutoGen / AG2 (autogen, register_function, GroupChat, GroupChatManager)
LLM	GPT-4o via OpenAI API; temperature 0 for deterministic, repeatable reasoning
Agents	AssistantAgent (x3) + UserProxyAgent (code execution disabled; human_input_mode = TERMINATE)
Tool binding	register_function() binds each Python tool to a calling agent with the proxy as executor
Orchestration	GroupChat with auto speaker selection, max 15 rounds, driven by GroupChatManager
Output	ChatResult object; full exchange available via tax_result.chat_history

9. Key Benefits

- **Separation of concerns** — each agent has a single, well-defined responsibility, making the system easy to extend and reason about.
- **Numeric reliability** — financial calculations live in deterministic Python tools, so figures are accurate, repeatable, and auditable rather than hallucinated.
- **Conversational orchestration** — the manager dynamically routes the conversation, so no rigid hard-coded pipeline is required.
- **Built-in compliance** — compliance validation surfaces missing PAN, Aadhaar, or bank details before filing, preventing rejected returns.
- **Reusability** — the same pattern generalises to other regulated, calculation-heavy domains (loans, payroll, insurance).

10. Limitations & Considerations

- **Statutory currency** — the tax slabs, rebate, and deduction ceilings are hard-coded for FY 2024-25 and must be updated each financial year.
- **Input parsing** — the scenario passes profile details as free text; richer flows would parse and validate structured inputs before computation.
- **Coverage** — the US tax path referenced in the design is a placeholder; only the India computation is implemented.
- **Advisory nature** — the assistant provides guidance and should not be treated as a substitute for a qualified tax professional for filing decisions.